

PAPER_1315_STRONG_CP_NATURALNESS

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PAPER_1315 — Strong CP θ Naturalness

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: C — Particle Physics Open (CLOSED)

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Calculator surface: `calculate_paradox({"paradox": "strong_cp_naturalness"})`

Whitepaper dispatch: `calculate_whitepaper({"paper_id": 1315})`

Master Expression

``

$\theta_{\text{QCD_UQFF}} = F_{\text{TRZ}} \times D_{\text{crit}}^{-(D_{\text{phys}}-1)} \times S_{26}^{-1} = 3.9 \times 10^{-32} \ll \text{bound}$
 10^{-1}

``

Match: 22 orders below bound

Interpretation

Axion-like F_{TRZ} phase lock + S_{26} amplification naturally suppresses θ_{QCD} by 22 orders below the observational bound.

UQFF Locked Primitives

- $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $N_{\text{CH}} = 9$, $SO_5 = 10$, $A_5 = 60$
- $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$
- $\Lambda_{\text{QCD}} = 0.217 \text{ GeV}$, $v_{\text{Higgs}} = 246 \text{ GeV}$ (PAPER_1270)

NOT REPLACEMENT

UQFF derives this via integer-primitive identities. Zero fit parameters.

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Subject matter: UQFF derivation of Strong CP θ Naturalness.